



Robótica Móvel e Inteligente / Intelligent Mobile Robotics
(Academic year of 2025-2026)

Assignment 2

**Robotic challenge solver
using the Webots simulation environment**

October, 2025

1 Objectives

This assignment requires each group to develop two robotic agents for controlling a simulated mobile robot. The goal is to successfully tackle two distinct robotic challenges, each demanding a different set of navigation skills.

The challenges included in this assignment are as follows:

- C2 - Mapping: The objective of this challenge is to explore an unknown maze to extract its map.
- C3 - Planning: The objective of this challenge is to explore an unknown maze in order to locate multiple target spots and compute the shortest closed path that allows to visit all of them, starting and ending at the starting spot.

Each of the challenges will be described in more detail later.

2 The Webots environment

The Webots simulation environment will be used to develop the agent to overcome the requested challenges. Webots is an open source robot simulator that is used to develop and test mobile robots in 3D virtual environments. It provides realistic physics, built-in robot models, and a variety of sensors and actuators. Robots in Webots are controlled by programs written in languages such as C, C++, Python, or Java, allowing the development and testing of navigation and control strategies without needing physical hardware.

The simulated robot (based on the e-puck, see figure 1) is a small, circular mobile robot equipped with two wheels (differential drive), 8 distance sensors, a color camera, 10 LEDs, a compass, and a GPS.

The available sensors depend on the challenge to be solved.

The simulated robot navigates in a delimited rectangular arena, 7 units tall and 14 units wide, with the unit length equal to 0.15 m. This navigable area can be seen as a bi-dimensional array of fixed-size cells, each cell being a square. A maze is defined by putting thin walls (0.02 m wide) between adjacent cells, which can be detected using the distance sensors. Each target spot has a unique identification color that is detectable by the camera sensor.

The Webots download page is:

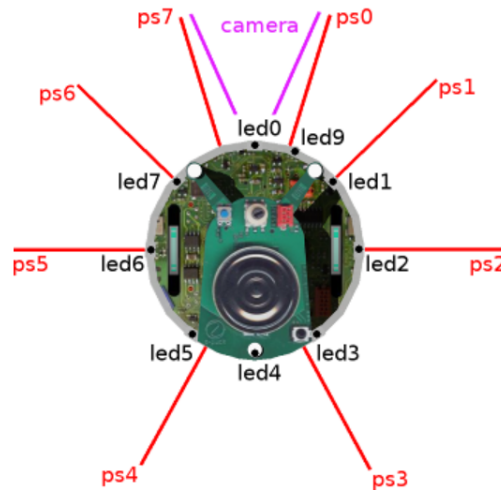


Figure 1: The simulated robot.

<https://cyberbotics.com/>

You should also clone and compile the **ciberRatoTools** repository (branch **cb_webots**) available at:

https://github.com/iris-ua/ciberRatoTools/tree/cb_webots

Instructions to compile the tools are available at the **README.md** file of the repository.

3 Challenges' description

3.1 C2 – Mapping challenge

As stated above, the objective of this challenge is to explore an unknown maze in order to extract its map.

Figure 2 shows an example of a maze suitable for this challenge, which will be available during development. At the beginning, the robot will be aligned with the X-axis (horizontal).

A GPS sensor and a compass are provided, both noiseless. Using these sensors, the agent can determine the robot's position and orientation in the maze at every cycle. Therefore, the problem of robot localization can be solved using these two sensors.

Navigation in the maze can be carried out using the approach adopted in Assignment 1. However, the task is more straightforward and yields better results when using the noiseless GPS and compass sensors. Distance sensors can then be used to identify the locations of walls and/or entrances throughout the maze.

To successfully complete this task, a strategy should be adopted that guarantees full exploration of the maze.

The maximum allowed simulation time for each maze is obtained by processing the **customData** string. For instance, in challenge C2 the format is: "FinalTime 350.0 NTargets 1", where 350.0 represents the maximum time in seconds, and NTargets represents the number of target spots (which is fixed at 1 for Challenge C2).

Collisions with walls will not be penalized but should always be avoided, as they may cause the agent to become stuck and consequently prevent it from completing the task.

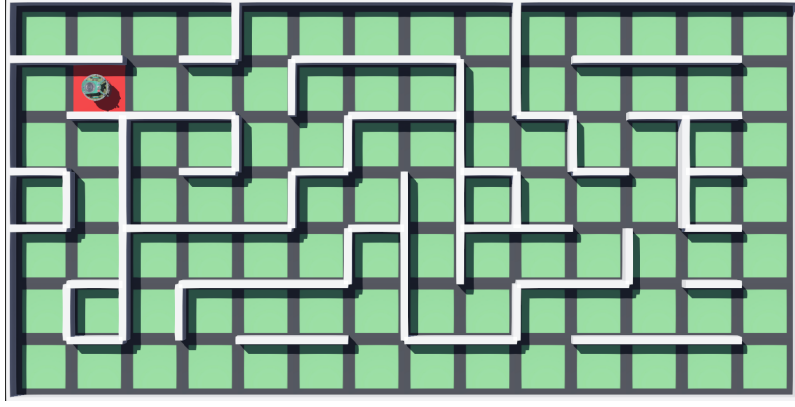


Figure 2: Example of a maze for challenge C2.

3.2 C3 – Planning challenge

The objective of this challenge is to explore an unknown maze to locate multiple target spots (between 3 and 7) and compute the shortest closed path that visits all target spots, starting and ending at the initial position. Each target spot is marked with a unique color, which can be detected by using the robot's camera. The mapping between the spot ID and the corresponding color is as follows: 0 – Red; 1 – Blue; 2 – Yellow; 3 – Magenta; 4 – Cyan; 5 – Orange; 6 – Black. Figure 3 shows an example of a maze suitable for this challenge, which will be available during development. At the beginning, the robot will be aligned with the X-axis (horizontal). Figure 4 presents another example of a maze, illustrating the unique colors used to mark each target spot.

As in Challenge 2, noiseless GPS and compass sensors are available, allowing the same navigation strategy to be applied in this challenge. The maze may not need to be explored in its entirety: exploration is only required until all target spots are located, and the agent can confirm that the shortest path has been determined.

The maximum allowed simulation time for each maze is obtained by processing the `customData` string. For instance, in challenge C3 the format is: "FinalTime 400.0 NTargets 3", where 400.0 represents the maximum time in seconds, and NTargets denotes the number of target spots, which, depending on the maze, ranges from 3 to 7.

Collisions with walls will not be penalized but should always be avoided, as they may cause the agent to become stuck and consequently prevent it from completing the task.

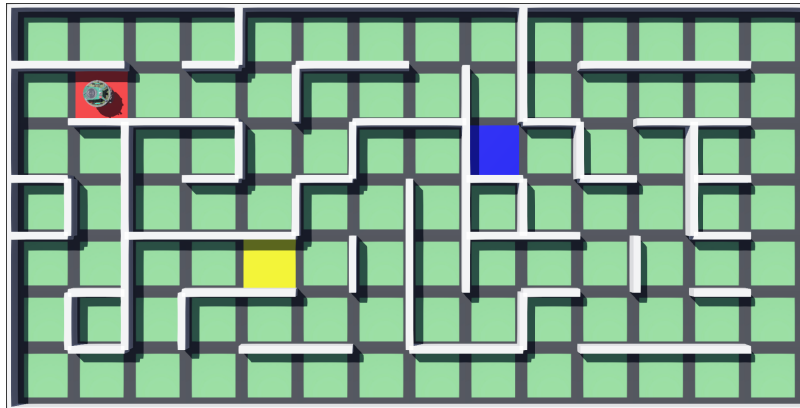


Figure 3: Example of a maze for challenge C3.

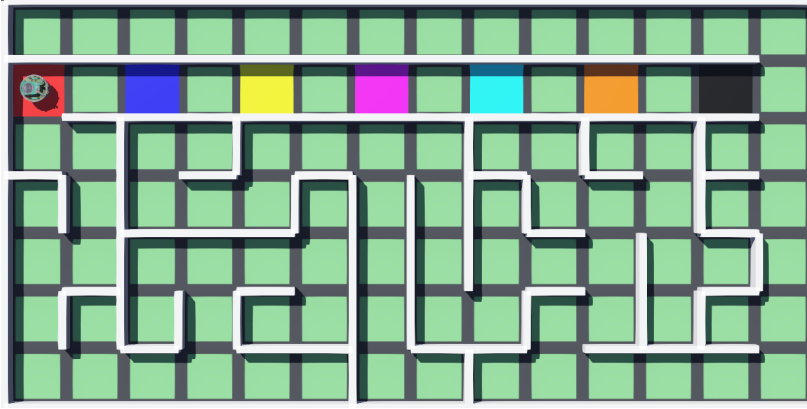


Figure 4: Colors used to identify each target spot.

4 Scoring

4.1 C2 – Mapping challenge

When a new simulation starts, the supervisor generates the file `mapping.out` which represents the map of the environment. This file is created in the `C2_supervisor` directory.

The agent should create a file, in the same format (see figure below for an example), that represents the robot's internal view of the maze map (this file should have the extension `.map`).

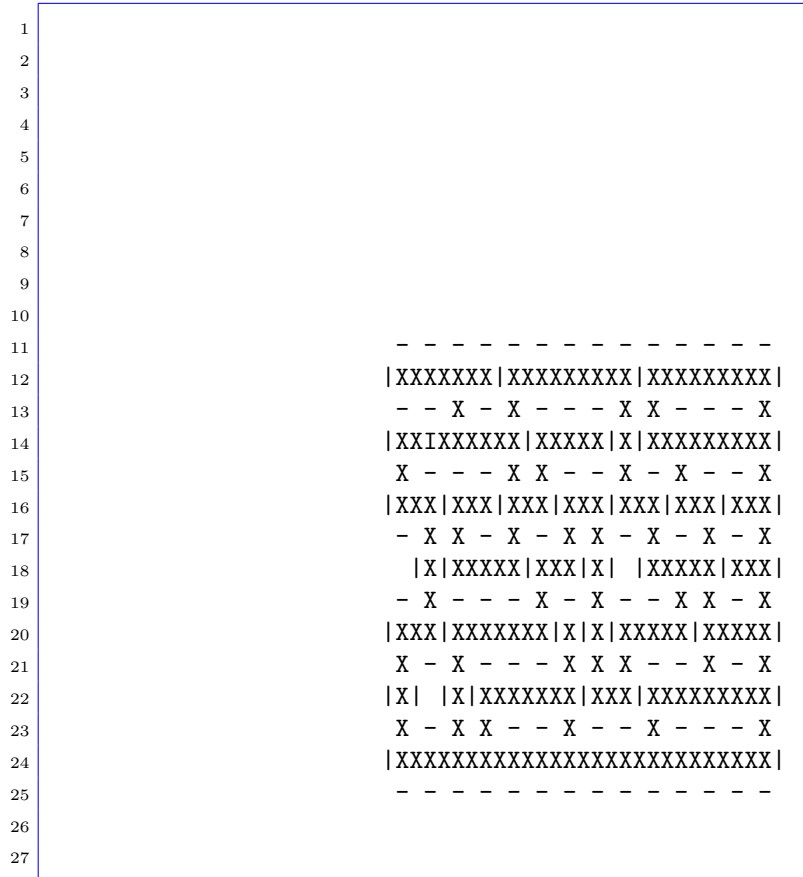
To verify the correctness of the generated map file, and to obtain the corresponding score, the `mapping_score.awk` script can be run, in the `C2_supervisor` directory, as follows:

```
gawk -f mapping_score.awk mapping.out «path-to-your-map-file»
```

To obtain the maximum achievable score for the current maze, the supervisor's output file can be compared against itself, as follows:

```
gawk -f mapping_score.awk mapping.out mapping.out
```

The map file is a text file that contains 27 lines, each with 55 characters. The center of the file always represents the starting position, marked as `I`. Below you can find an example of a map file.



The final assessment will be based on the map score and on the time taken to finish the challenge.

4.2 C3 – Planning challenge

When the agent starts, the simulator generates the file **planning.out**, which includes the map of the environment and additional information on the best path to visit all targets.

Each time the agent executes, it should create a file that specifies the path to go from the starting position, visit all targets, and return to the starting position. This file should have the extension **.path**.

The path file is a text file that contains, in each line, the x and y coordinates of the center of the cells that are visited by the path. Coordinates are relative to the starting position and are measured in cell units. The starting and ending coordinates should always be 0 0.

Below is an example of a path file, presented in three columns to reduce page space.

The correctness of the path file can be checked by executing the **planning_score.awk** script, in the **C3_supervisor** directory, as follows:

```
gawk -f planning_score.awk planning.out «path-to-your-path-file»
```

The highest possible score is 1.0.

1	0	0	# 0	15	9	-3	29	1	-5
2	1	0		16	9	-4	30	0	-5
3	2	0		17	8	-4	31	-1	-5
4	3	0		18	8	-5	32	-1	-4
5	3	1		19	7	-5	33	-1	-3
6	4	1		20	6	-5	34	0	-3
7	5	1		21	5	-5	35	0	-2
8	6	1		22	5	-4	36	0	-1
9	7	1		23	4	-4	37	-1	-1
10	7	0		24	4	-3	38	-1	0
11	7	-1	# 1	25	3	-3	# 2	39	0 0 # 0
12	8	-1		26	2	-3		40	
13	8	-2		27	1	-3			
14	9	-2		28	1	-4			

The final assessment will be based on the path score and on the time taken to finish the challenge.

5 Assessment

The assessment of this assignment will be composed of 2 components: performance of the developed agents and a presentation.

- The agents will be tested and graded by teachers using different maze configurations, in batch mode, in their own computers. Thus, the format specified for their execution is mandatory.
- Each group will make a presentation of its work, consisting of an oral presentation, based on pdf slides, and a short discussion of the work (10 minutes maximum for presentation).

6 Deliverables and deadline

- Source code of the developed agents.
 - The code should include 2 folders called `agent_nmec1_nmec2_c2` and `agent_nmec1_nmec2_c3`, regardless of the programming language used (*nmec1* and *nmec2* should be replaced by your UA id number). `agent_nmec1_nmec2_c2` should include the agent that solves the mapping challenge (C2) and `agent_nmec1_nmec2_c3` should include the agent that solves the planning challenge (C3).
 - agent folders should both contain a script file called `setup.sh` that will be executed by the bash that runs webots, before executing webots. This script should, for example, compile the agent if it is written in C, or create and activate a virtual environment if the agent is written in Python and uses a virtual environment. Even if the code is developed in Python and does not use a virtual environment, the script should exist (it may do nothing).
 - **Source code must be submitted to the Moodle RMI website by November, 9.**
- Presentation covering both Assignment 1 and Assignment 2 (in PDF format).
 - The presentation should include:
 - * Initial slide (names, nmecs, course, assignment, etc.).
 - * For each task (control, mapping, planning) you should present your **approach** for solving each (sub)problem, the experiments carried out and the **results** obtained (use the slides you deem appropriate).
 - * Final slide with the conclusions of the work.

- * Slide with bibliography/sources used in the work (this slide does not need to be presented).
- All slides should be numbered.
- The presentation time is limited to 10 minutes.
- **Presentation must be submitted to the Moodle RMI website by November, 12.**

7 Bibliography

- “Principles of Robot Motion: Theory, Algorithms, and Implementations”, Howie Choset et al., MIT Press, Boston, 2005.
- “Introduction to Autonomous Mobile Robots”, Second Edition, Roland Siegwart et al., MIT Press, 2011.
- “Artificial Intelligence: A Modern Approach”, 4th edition, Russel and Norvig, Pearson, 2021.
- “Webots User Guide”, <https://cyberbotics.com/doc/guide/index>.